NASA SPACE APPS 2026 · DANCING WITH THE SARS

Undula

Hear the ground move.

NASA-ISRO **NISAR** radar, turned into something anyone can see, hear and understand.

TEAM **SHIBLANGA** · EGYPT



THE PROBLEM

The ground under more than 40 million people is moving.

The Nile Delta is sinking while the sea rises. Too slowly to see, fast enough to matter.

11 mm/yr
The Delta sinking, up to
Damietta

3.4 mm/yr
The sea rising
Mediterranean

40%+
of Egyptians live on the Delta

Sinking and sea-rise rates: Hassan et al., 2025.



THE EYE IN THE SKY

NISAR feels the ground move.

NASA and ISRO radar satellite
A 12-metre antenna, L-band and S-band radar.

Every 12 days
Images nearly all of Earth's land and ice.

Day and night, through clouds
Radar brings its own light.



Two passes, compared
Each rainbow cycle is about **12 cm** of motion toward or away from the satellite.



NISAR · SUN-SYNCHRONOUS ORBIT

AROUND THE WORLD

Ten real places. Three ways the ground moves.



- **5** Sinking ground: cities, a delta, a farm valley
- **3** Volcanoes swelling
- **2** Faults tearing and creeping

01 SEE

Real radar, made readable.

Ten real places
Cities sinking, volcanoes swelling, faults tearing. Each with its story.

Watch the rings grow
Every ring is the land moving, pass after pass.

Real NISAR data
Actual radar passes from 2026, played like a film.

Any spot on Earth
NISAR's latest image of your own street, live from NASA.





02 HEAR & IMAGINE

Hear the ground. See 2100.



Every centimetre plays a note.

Falling for sinking, rising for uplift.

UNDER THE SEA BY 2100, WITH THE SINKING

3,491 km²

SEA RISE ALONE

1,557 km²

More than half of it is the land sinking.

Rough estimate at constant 2026 rates, Copernicus DEM GLO-90.



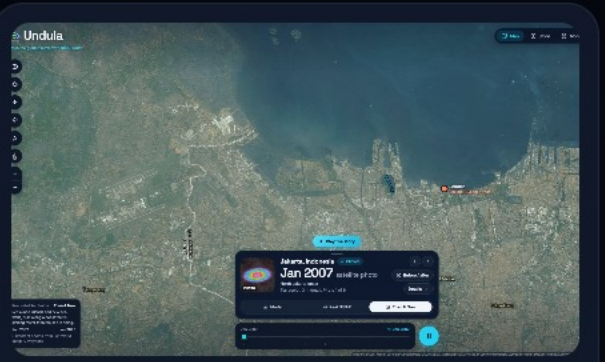
03 EXPLORE & TRUST

Walk it. Compare it. Check it.



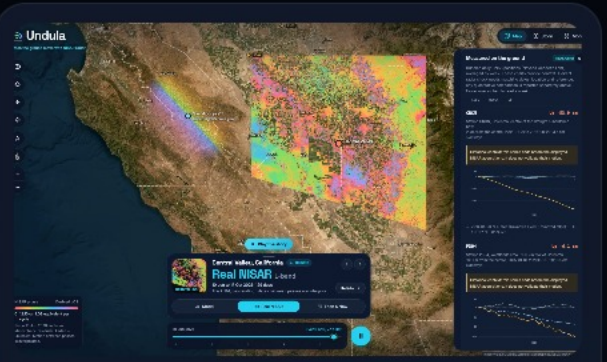
Explore as Astrounit

Walk the map and discover each region's story.



Then & now

Real satellite photos across the years.



GPS on the ground

Station records beside the radar, with honest notes on what they can and can't confirm.

MEASURED FROM THE FULL NISAR DATA

The Delta's farming season, seen by radar

10 GCOV dates · HH + HV · June to October 2026

Independent Landsat check: **8** date pairs · HV/NDVI rank correlation **0.67** · not crop classification



BUILT ON OPEN SPACE DATA

NASA/JPL-Caltech & ISRO NISAR via ASF · NASA Earthdata (CMR) & Search · Nevada Geodetic Lab GPS · Copernicus DEM GLO-90 · Esri World Imagery & Wayback



ALL OF IT, IN ONE APP

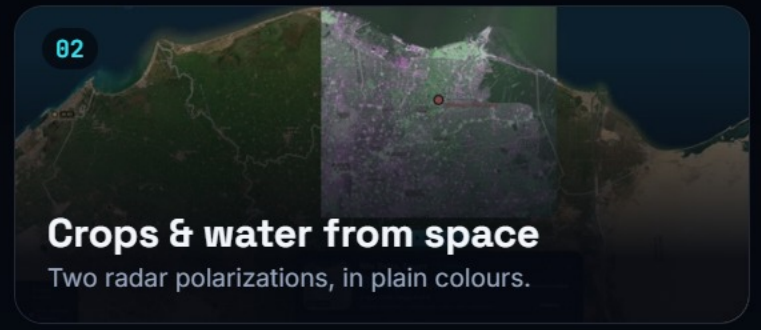
Nine ways to feel the ground move.



01

Real NISAR, played like a film

Actual 2026 radar passes, pass by pass.



02

Crops & water from space

Two radar polarizations, in plain colours.



03

Any spot on Earth, live

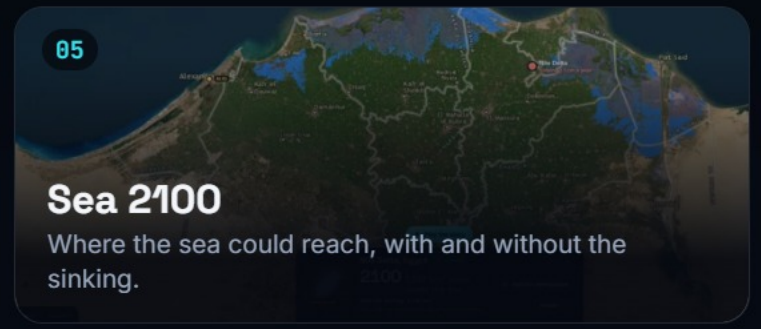
NISAR's latest image, straight from NASA.



04

Hear the ground

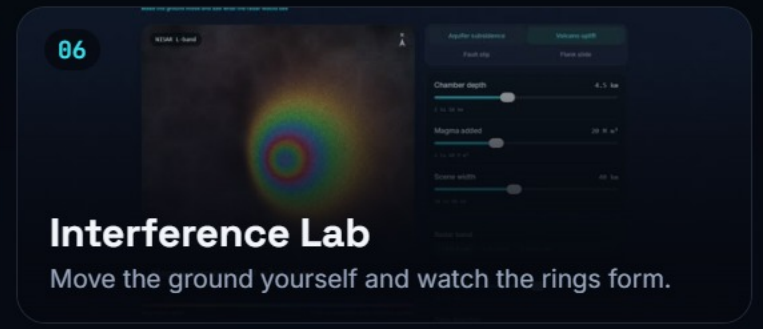
Every centimetre of motion plays a note.



05

Sea 2100

Where the sea could reach, with and without the sinking.



06

Interference Lab

Move the ground yourself and watch the rings form.



07

Explore as Astrounit

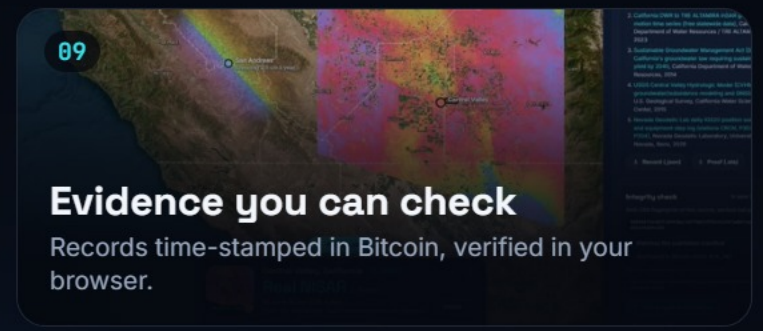
Walk the map and discover each region's story.



08

Then & Now

Real satellite photos across the years.



09

Evidence you can check

Records time-stamped in Bitcoin, verified in your browser.

WHY IT MATTERS

From a satellite 747 km up, to a farmer's field in the Delta.

- **For everyone.** No science degree needed: see it, hear it, understand why.
- **Honest by design.** Every number shows its source. Estimates say they're estimates.
- **Grows with NISAR.** Every 12-day pass adds new places and new chapters.

Try Undula

github.com/Mohamed06/Shibanga/tree/undula

Undula. Hear the ground move.

TEAM SHIBLANGA · OMAR · MOHAMED · ISLAM

